

RUST PROGRAMMING

Structures

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STRUCTURES

- Types of structures
 - Name-Fields
 - Tuple-Like
 - Uni-Like
- Defining methods with “impl”
- Associated Consts
- Generic Structs
- Structs with Lifetime Parameters
- Deriving Common Traits for Struct Types
- Interior Mutability

Named-Field Structures: Concept Overview

1. Structures in C/C++ are simple containers for data, with members public by default in C but configurable in C++.
2. Rust structures are more powerful, supporting features like ownership and borrowing.
3. Rust's structs don't include methods or access control inherently but gain them through impl blocks

```
struct MyStruct {  
    x: i32,           // Stack  
    y: String,        // Heap (the String data, not the pointer/length/capacity)  
    z: Box<i64>,       // Heap (the i64 inside the Box)  
}  
  
fn main() {  
    let a: i32 = 10;           // Stack  
    let b = String::from("hi"); // Heap  
    let c = Box::new(20);      // Heap  
  
    let my_struct: MyStruct = MyStruct {  
        x: 5,                 // Stack  
        y: String::from("Rust"), // Heap  
        z: Box::new(100),      // Heap  
    };  
}
```


Named-Field Structures: Basics

```
1 #[derive(Debug)]
  struct FriendNode {
    name: String,
    age: i32,
    friends: Vec<FriendNode>,
  }

  fn main() {
2   let mut david = FriendNode {
        age: 25,
        name: "Kamal".to_string(),
        friends: Vec::new();
    println!("{:?}", david);

3   david.name = "David Paul".to_string();
    println!("{:?}", david.name);
  }
```

Output

```
./struct_rc
FriendNode { name: "Kamal", age: 25, friends: [] }
"David Paul"
```

1. Defining structure

- Field name and type are separated by `:`
- Each field is separated `,`
- No `,` at the end of structure definition

2. Creating an instance and initializing

- Mention field name along with value
- All the fields should be initialized
- Order does not matter

3. Modifying and accessing members

- Use `.` to access or modify

Named-Field Structures: Copy

```
fn main() {  
    let david = FriendNode {  
        age: 25,  
        name: "Kamal".to_string(),  
        friends: Vec::new();  
    }  
  
    let john = FriendNode {  
        name: "John".to_string(),  
        .. david}; // Copy partially from david to john  
    println!("john details: {:?}", john);  
  
    let peter = john; // Clone david to peter  
    println!("peter details: {:?}", peter);  
    // error[E0382]: borrow of partially moved value: `david`  
    //println!("david details: {:?}", david);  
    // error[E0382]: borrow of moved value: `john`  
    //println!("john details: {:?}", john);  
}
```

Output

```
% ./struct_basic1  
john details: FriendNode { name: "John", age: 25, friends: [] }  
peter details: FriendNode { name: "John", age: 25, friends: [] }
```

1. Copying data from one to another instance using **.. EXPR**
2. Partially copied instance (david) loses the ownership

Named-Field Structures:

```
fn main() {  
    let david: FriendNode = FriendNode {  
        age: 25,  
        name: "David".to_string(),  
        friends: Vec::new();  
  
    let john: FriendNode = FriendNode {  
        friends: Vec::new(),  
        name: "John".to_string(),  
        .. david}; // Copy david to john  
    println!("john details: {:?}", john);  
  
    let peter: FriendNode = john;  
    println!("peter details: {:?}", peter);  
    // No ERROR :)  
    println!("david details: {:?}", david);  
}
```

Output

```
% ./struct_basic1  
john details: FriendNode { name: "John", age: 25, friends: [] }  
peter details: FriendNode { name: "John", age: 25, friends: [] }  
david details: FriendNode { name: "Kamal", age: 25, friends: [] }
```

CHALLENGE

1. No ERROR: Why?

Defining methods with “impl”:

```
#[derive(Debug)]
1 implementation
2 struct IdsEvent {
    id: u32,
    event: String,
}

1 impl IdsEvent {
2     fn new(id: u32, event: &str) -> IdsEvent {
        if event.len() == 0 {
            return IdsEvent {
                id: id,
                event: "No Event".to_string(),
            };
        } else {
            return IdsEvent {
                id: id,
                event: event.to_string(),
            };
        }
    }

3     fn get_id(&self) -> u32 {
        self.id
    }
}

4 fn main() {
    let mut ids_event1: IdsEvent = IdsEvent::new(id: 100, event: "");
    println!("ids_event1: {:?}", ids_event1);
    println!("ids_event1 id: {}", ids_event1.get_id());
}
```

Output

```
% ./struct_impl
ids_event1: IdsEvent { id: 100, event: "No Event" }
ids_event1 id: 100
```

1. Implementing member function for structures using **impl**
2. **new** function called as **Associated Functions** which does not need “self”
3. Writing a member function, called on instance
4. Invoking “new” associated function to construct instance

“impl” methods:



Methods in “impl” can be categorized two ways:

- Instance methods:
 - **Consumable Instance methods:** Which demands `self` (depends on definition, no default behavior)
 - **Mutable Instance methods:** Which demands `&mut self`
 - **Instance methods:** Which demands `&self`
 - Ex: `get_id`
- Associated functions: Functions that are defined within an `impl` block for a struct but don't take `self` as a parameter.
 - Ex: `new`

Example with all types of functions in “impl”:

```
struct Rectangle {
    width: u32,
    height: u32,
}

impl Rectangle {
    // Immutable borrowing: the method borrows `self` immutably.
    pub fn area(&self) -> u32 {
        self.width * self.height
    }

    // Mutable borrowing: the method borrows `self` mutably, allowing modification.
    pub fn scale(&mut self, factor: u32) {
        self.width *= factor;
        self.height *= factor;
    }

    // Taking ownership: the method takes ownership of `self`.
    pub fn destroy(self) {
        println!("Rectangle with dimensions {}x{} is destroyed.", self.width, self.height);
        // The instance is consumed here and can't be used afterward.
    }

    // Associated function: does not take `self`.
    pub fn new(width: u32, height: u32) -> Self {
        Self { width, height }
    }
}
```

```
fn main() {
    // Using the associated function `new` to create an instance of `Rectangle`.
    let mut rect: Rectangle = Rectangle::new(width: 30, height: 50);

    // Immutable borrowing: calling the `area` method, which doesn't modify `rect`.
    println!("The area of the rectangle is {} square pixels.", rect.area());

    // Mutable borrowing: calling the `scale` method, which modifies `rect`.
    rect.scale(factor: 2);
    println!("After scaling, the area is {} square pixels.", rect.area());

    // Taking ownership: calling the `destroy` method, which consumes `rect`.
    rect.destroy();
    // At this point, `rect` can no longer be used because it has been consumed.
}
```

Output

```
% ./all_types_of_methods
The area of the rectangle is 1500 square pixels.
After scaling, the area is 6000 square pixels.
Rectangle with dimensions 60x100 is destroyed.
```

General associative methods used:



- **new**

- Creates and returns a new instance of the struct.
- Example: `MyStruct::new()`.

- **default**

- Returns a default instance of the struct. Often used when implementing the Default trait.
- Example: `MyStruct::default()`.

- **from / from_***

- Converts from another type to the struct. Often used for custom conversions.
- Example: `MyStruct::from("string")`.

- **with_***

- Alternative constructors that create an instance with specific fields initialized.
- Example: `MyStruct::with_default_values()`.

- **create_***

- Factory methods that generate instances, sometimes in complex or multiple-step scenarios.
- Example: `MyStruct::create_pair(1, 2)`.

- **max_value / min_value**

- Returns the maximum or minimum value that a struct or its fields can hold.
- Example: `MyStruct::max_value()`.

- **description**

- Returns a static description or string related to the struct.
- Example: `MyStruct::description()`.

- **from_str**

- Converts a string to an instance of the struct, commonly used for parsing.
- Example: `MyStruct::from_str("input string")`.

- **from_tuple**

- Converts a tuple to an instance of the struct.
- Example: `MyStruct::from_tuple((1, 2))`.

- **from_*_slice**

- Converts a slice of a specific type to an instance of the struct.
- Example: `MyStruct::from_u8_slice(&[1, 2, 3])`.

Cont...



- **constants**

- Associated constants that represent fixed values associated with the struct.
- Example: `MyStruct::MAX_VALUE`.

- **is_valid**

- Returns a boolean indicating whether some condition is met for a type or value.
- Example: `MyStruct::is_valid(some_value)`.

- **parse**

- Parses a value into the struct, often used for implementing the `FromStr` trait.
- Example: `MyStruct::parse("input")`.

- **get_instance**

- Returns a singleton or globally accessible instance of the struct.
- Example: `MyStruct::get_instance()`.

- **to_string**

- Converts the struct to a string representation. Often used with the `ToString` trait.
- Example: `MyStruct::to_string()`.

- **to_bytes / from_bytes**

- Converts the struct to or from a byte array, useful for serialization or binary formats.
- Example: `MyStruct::to_bytes()`.

- **zero**

- Returns a zeroed-out instance of the struct, often used in numeric contexts.
- Example: `MyStruct::zero()`.

- **unit**

- Returns a unit instance (for unit-like structs or types that represent an empty state).
- Example: `MyStruct::unit()`.

- **init**

- Initializes some internal state or system related to the struct.
- Example: `MyStruct::init()`.

- **configure**

- Configures some settings or parameters related to the struct.
- Example: `MyStruct::configure(settings)`.

Cont...

- **all**
 - Returns a collection of all possible instances, often used with enums or specific struct types.
 - Example: `MyEnum::all()`.
- **empty**
 - Returns an empty instance of a struct, often used for collections or containers.
 - Example: `MyStruct::empty()`.
- **from_iter**
 - Creates an instance of the struct from an iterator, often used for collections.
 - Example: `MyStruct::from_iter(iterator)`.
- **len / is_empty**
 - Returns the length or checks if the struct is empty, typically used for collection-like structs.
 - Example: `MyStruct::len()` or `MyStruct::is_empty()`.
- **default_instance**
 - Returns a predefined default instance that might be used frequently.
 - Example: `MyStruct::default_instance()`.

Tuple-Like Structures: Basics and “impl”

```
#[derive(Debug)]
```

```
1 implementation
```

1

```
struct Color(u8, u8, u8);
```

```
impl Color {
```

```
    fn new(r: u8, g: u8, b: u8) -> Color {  
        Color(r, g, b)  
    }
```

```
    fn print_color (&self) {  
        println!("Color values: {}, {}, {}", self.0, self.1, self.2);  
    }
```

```
}
```

```
fn main() {
```

2

```
    let black: Color = Color(0, 0, 0);
```

```
    let mut white: Color = Color(255, 255, 255);
```

3

```
    println!("Black color values: {}", black.0);
```

```
    white.0 = 0;
```

```
    white.print_color();
```

```
}
```

Output

```
% ./tuple_struct_basic  
Black color values: 0  
Color values: 0, 255, 255
```

1. Defining structure

- No need of fields names, only types are enough.

2. Creating instance

1. Order matters

3. Modifying and accessing members

- Use **.index** to access or modify

Misc: Structures

```
use std::mem;
```

```
#[derive(Debug)]
```

```
0 implementations
```

```
struct IdsEvent {
```

```
    pub sensor_id: u16,
```

```
    pub id: u32,
```

```
    pub severity: u8,
```

```
    pub timestamp: u64,
```

```
    pub data: [u8; 10],
```

```
}
```

```
#[repr(C)]
```

```
#[derive(Debug)]
```

```
0 implementations
```

```
struct IdsEvent2 {
```

```
    pub sensor_id: u16,
```

```
    pub id: u32,
```

```
    pub severity: u8,
```

```
    pub timestamp: u64,
```

```
    pub data: [u8; 10],
```

```
}
```

```
fn main() {
```

```
    let event: IdsEvent = IdsEvent {
```

```
        sensor_id: 0,
```

```
        id: 1,
```

```
        severity: 2,
```

```
        timestamp: 3,
```

```
        data: [4; 10],
```

```
    };
```

```
    println!("Size of IdsEvent: {}", mem::size_of::<IdsEvent>());
```

```
    println!("Size of IdsEvent2: {}", mem::size_of::<IdsEvent2>());
```

```
}
```

Output

```
% ./strucut_padding
Size of IdsEvent: 32
Size of IdsEvent2: 40
```

Rust does not promise the order of fields in memory.

1. The **`#[repr(C)]`** attribute in Rust is used to control the memory layout of data structures, making them compatible with the C programming language's memory layout.

Important when you're interfacing with C libraries.

Structures: Default traits

```
1 #[derive(Debug, Clone, Copy, PartialEq, Eq)]  
  0 implementations  
2 struct Point {  
3     x: i32,  
4     y: i32,  
5 }  
6  
7 fn main() {  
8     let point: Point = Point { x: 10, y: 20 };  
9     let mut point2 = point.clone();  
10    let mut point3: Point = point;  
11  
12    point2.x = 11;  
13    point3.x = 12;  
14    println!("{:?}", point);  
15    println!("{:?}", point2);  
16    println!("{:?}", point3);  
17  
18    println!("Eq = {}", point == point2);  
19 }
```

- In Rust, traits define shared behavior that types can implement.
- Deriving common traits is often needed when you want to use standard behaviours like equality checks, debugging, or cloning without having to manually implement them.

1. Rust has several built-in traits that can be derived automatically using the `#[derive]` attribute.

Defining a simple trait:

```
1 trait Greet {  
2     | fn greet(&self) -> String;  
3 }  
4  
5 1 implementation  
5 struct Person {  
6     | name: String,  
7 }  
8  
9 2  
10 impl Greet for Person {  
11     | fn greet(&self) -> String {  
12         | format!("Hello, my name is {}!", self.name)  
13     }  
14 }  
15  
16 fn main() {  
17     | let person: Person = Person {  
18         | name: String::from("Alice"),  
19     };  
20     | println!("{}", person.greet());  
21 }
```

1. Define own trait
2. Implementing for Person

Defining a simple trait: Trait functions

```
fn say_hello<T: Greet>(entity: &T) {  
    println!("{}", entity.greet());  
}
```

1. “say_hello” can be called for any structure which has implemented trait “Greet”

Thanks